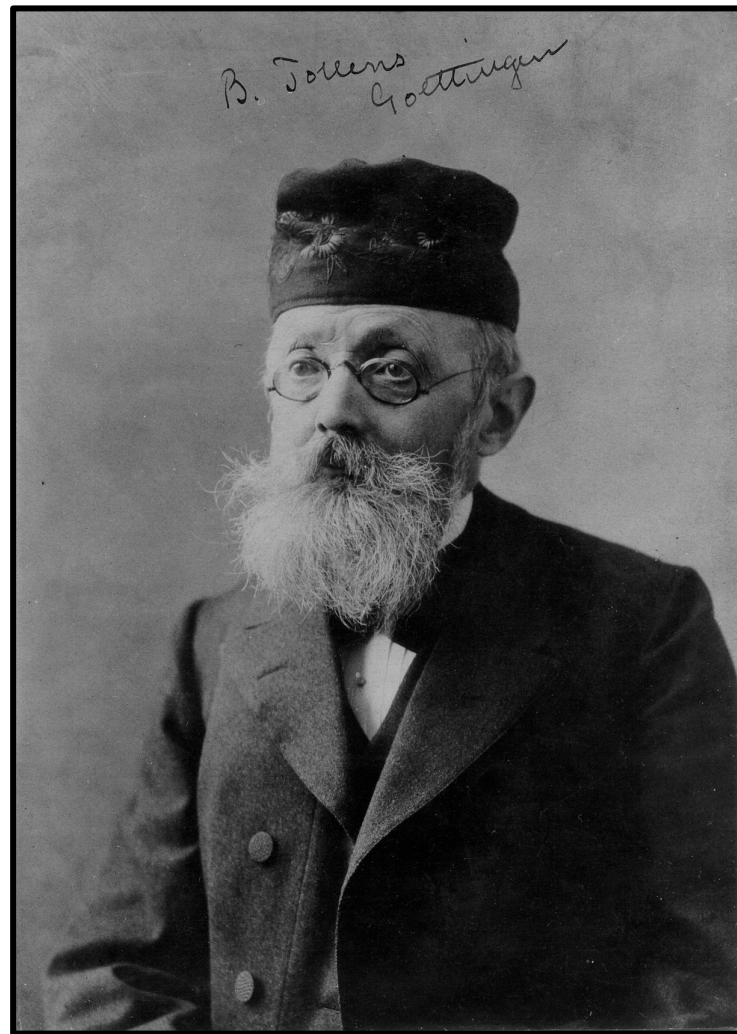
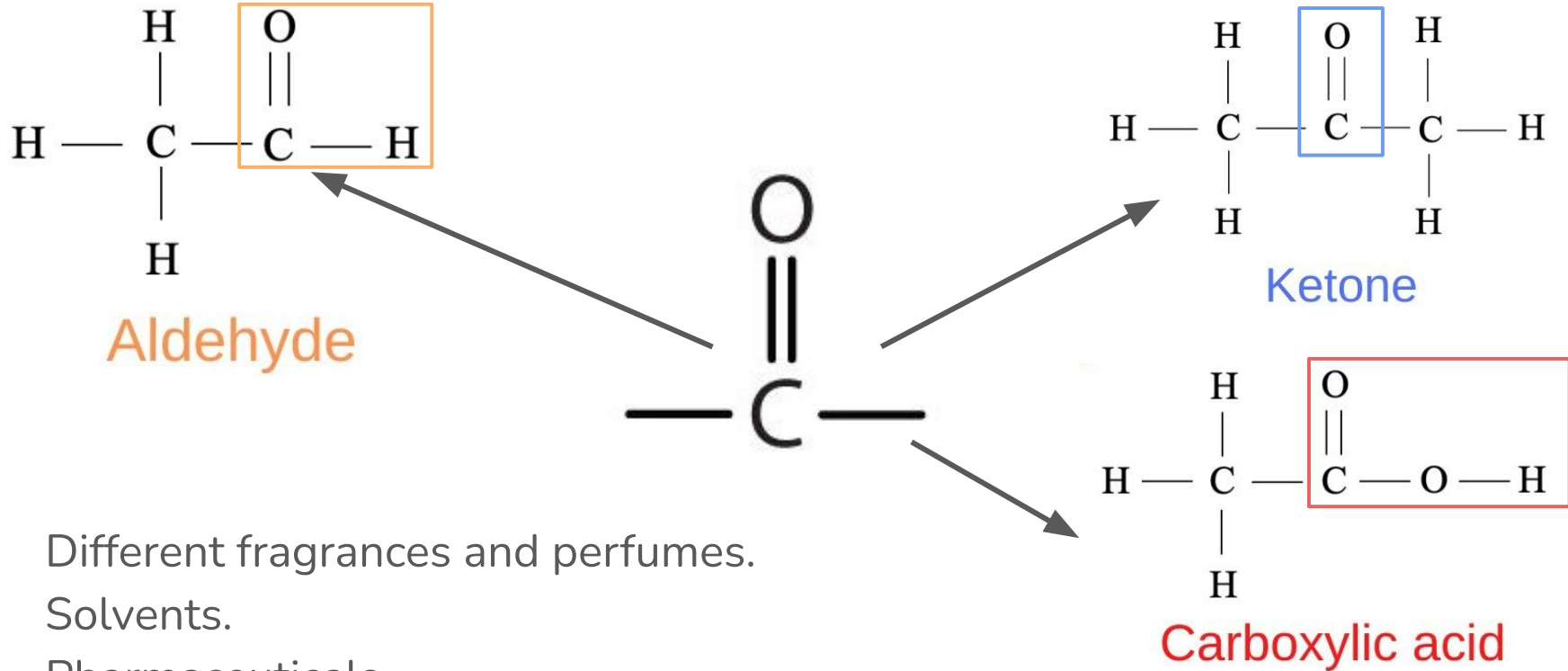


Tollen's Reagent

Testing for an
aldehyde.



What is the Carbonyl Group?

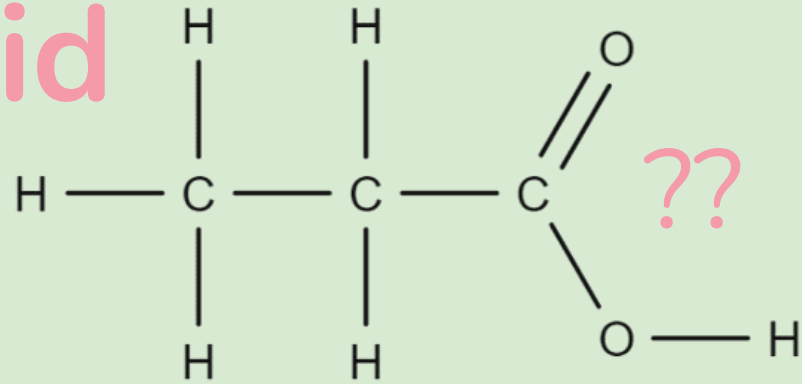


- ★ Different fragrances and perfumes.
- ★ Solvents.
- ★ Pharmaceuticals.
- ★ **Retinal** (this is why you can see).

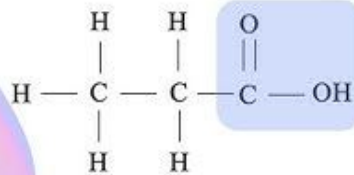
Let's
PLAY

Aldehyde,
Carboxylic Acid or
Ketone?

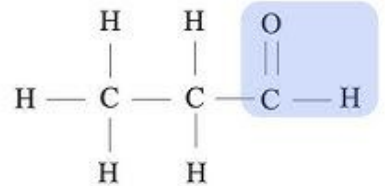
Carboxylic acid



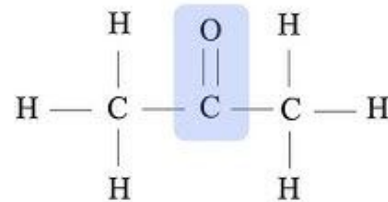
Carboxylic Acid



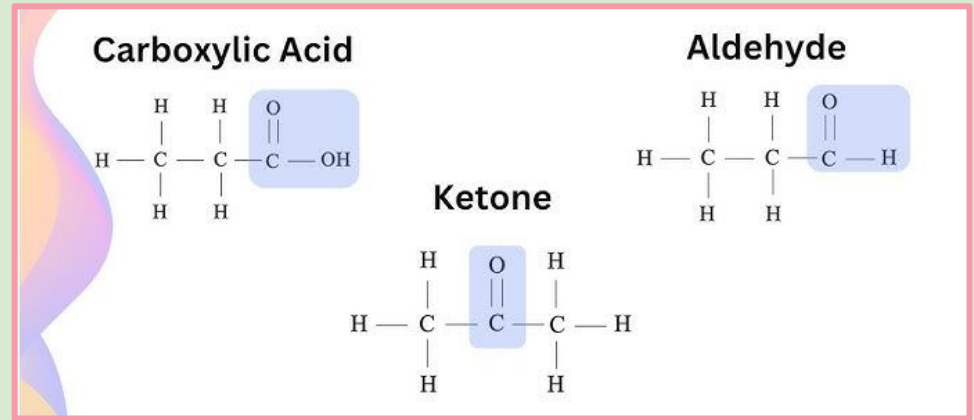
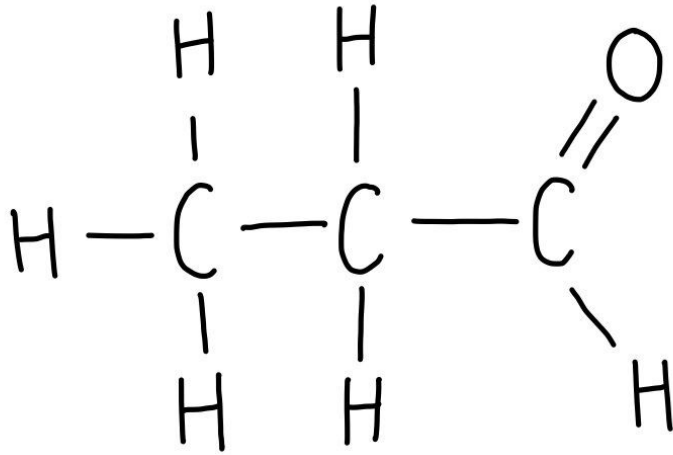
Aldehyde



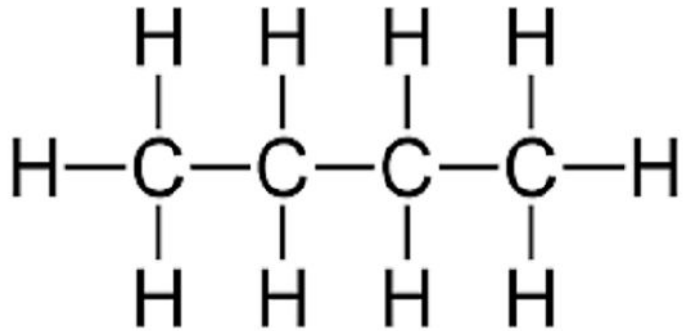
Ketone



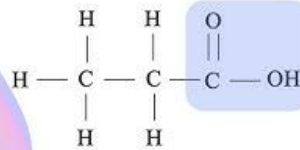
Aldehyde



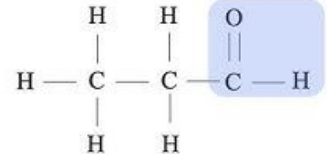
Nothing - it
was a trick!!



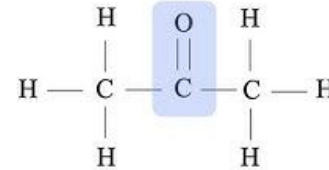
Carboxylic Acid



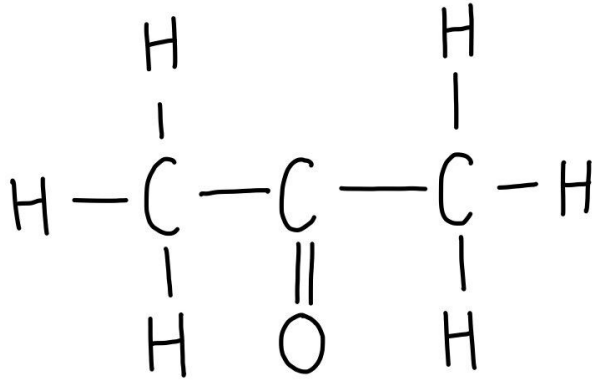
Aldehyde



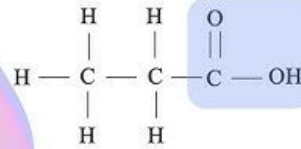
Ketone



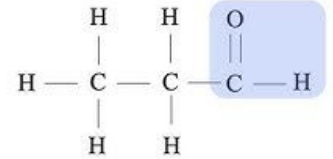
Ketone



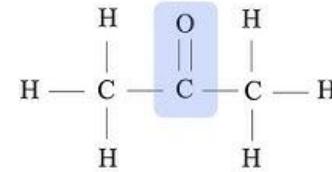
Carboxylic Acid



Aldehyde

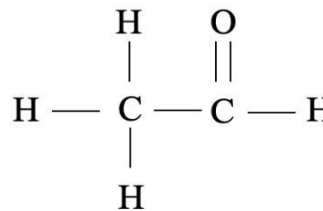
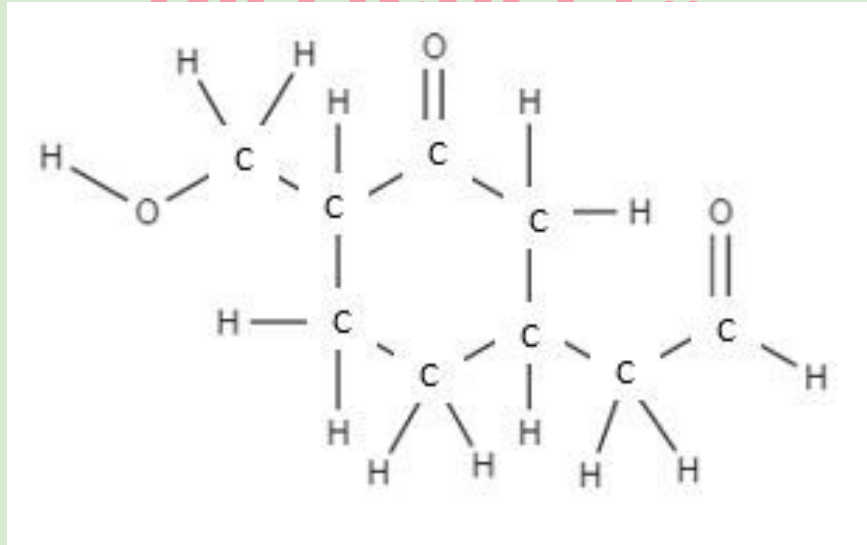


Ketone

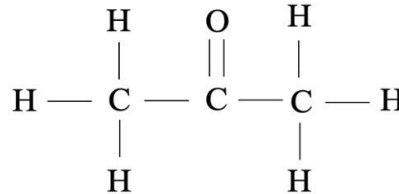


EXTRA, SUPER DUPER HARD DIFFICULTY!!

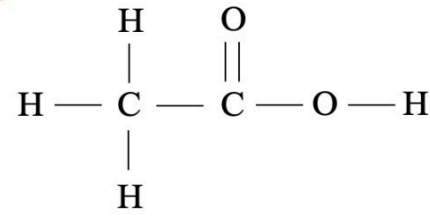
Ketone and
Aldehyde!!



Aldehyde



Ketone



Carboxylic acid

Method

1. Add 3-4 drops sodium hydroxide to 2cm^3 silver nitrate.
2. Add dilute ammonia solution to the test tube until the brown precipitate disappears
3. Add 10 drops of the sample
4. Use a water bath to warm.



What is happening?

- ★ This is the solution we made; by adding ammonia and silver nitrate.



- ★ When we add the **aldehyde** it undergoes a special type of reaction.
 - This is an oxidation reaction.
 - Oxygen is added to the aldehyde.
- ★ When this happens silver is formed on its own and forms the mirror on the test tube.

